

Drug Delivery

Nanoparticle Biodistribution Analysis

The rapid growth of nanotechnology-based drug delivery has made nanoparticles one of the most promising approaches for cancer treatment. This project analyzes nanoparticle biodistribution to investigate how different nanoparticle properties influence drug distribution across tumors and healthy organs. The insights gained help optimize nanoparticle design, improve tumor targeting efficiency, and reduce off-target toxicity.

Track: **Data Analysis** | Initiative: **DEPI** | Date: **11 July 2026** | Supervisor: **Eng. Mohamed Alemam**

Team Member

Aalaa Moataz Ahmed ElSayed

Team Leader

Shahd Fekry Mohammed Shawky

Team Member

Mohand Yasser Hussein Hanfy

Team Member

Mohamed Mostafa Ahmed Mahmoud

Team Member

Youssef Mohamed Saber Zedan

Team Member

Yahya Abdel Nasser Abdel Naim

Project Idea

Problem Statement

Traditional drug delivery distributes drugs throughout the body, causing side effects and low treatment efficiency. This project investigates how nanoparticle properties influence tumor targeting to optimize and develop more effective nanoparticle formulations with higher tumor uptake and lower toxicity.

Proposed Solution

Develop a complete data analytics solution that analyzes nanoparticle biodistribution experiments using:

Data Engineering Pipeline

Exploratory Data Analysis

Interactive Power BI Dashboard

AI Model: Hyperparameter Fine-Tuned Multi-Output XGBoost

Simultaneously predicts Tumor_%ID and Selectivity_Index using nanoparticle design features, enabling rapid evaluation of new nanoparticle formulations.

Unique Value

Complete End-to-End Data Pipeline: Seamless engineering from raw logs to deployment.

Medallion Architecture (Bronze → Silver → Gold): Governed and highly secure step-by-step transformation.

Star Schema Data Model: Optimized for high-speed calculation and reporting.

Interactive Power BI Dashboard: Rich interactive panels designed specifically for research use.

Machine Learning using Multi-Output XGBoost: Predicts nanoparticle performance classes using hyperparameter fine-tuning and real-time deployment via Streamlit.

Actionable Insights: Translates complex, abstract mathematical metrics into guidance for biomedical researchers.

Project Wireframe

The diagnostic interface design structure of the interactive command dashboard. Below is an overview of the planned sheets and their respective analytical roles.

Dashboard Page	Description
Home	Project overview, KPIs, and navigation.
Overview	General dataset statistics and biodistribution summary.
Nanoparticle Design	Size, Shape, Core Material Analysis.
Targeting Strategy	Active vs Passive Targeting Performance.
Tumor Biology	Tumor Sites, Cell Lines, Organ Distribution.
Surface Coating (PEG)	PEGylation and Shell Material Analysis.
Time & Dosage	Time-dependent Biodistribution Analysis.
Responsive Release	Release Mechanism Comparison.
Organic vs Inorganic	Nanoparticle Class Comparison.
Machine Learning	Prediction Results & Model Performance.

End Users + Features

Target Users

Biomedical Researchers: Evaluating target efficiency and organ retention curves.

Pharmaceutical Companies: Investigating scaling potentials and structural options.

Drug Delivery Scientists: Seeking optimized delivery variables for laboratory trials.

Oncology Researchers: Correlating cell line responses with treatment rates.

Data Analysts in Healthcare: Consuming cleaned, modeled datasets for secondary analytics.

Key Features

Interactive Dashboard: Clean visual grids.

Dynamic Slicers: Seamless cross-filtering.

KPI Monitoring: Real-time success metrics.

Biodistribution: Deep spatial analysis.

Targeting Strategy: Passive vs Active.

Performance: Material efficiency stats.

Time-Series: Multi-interval tracking.

Hyperparameter Fine-Tuned AI Classification: Predicts Tumor Uptake and Selectivity performance classes.

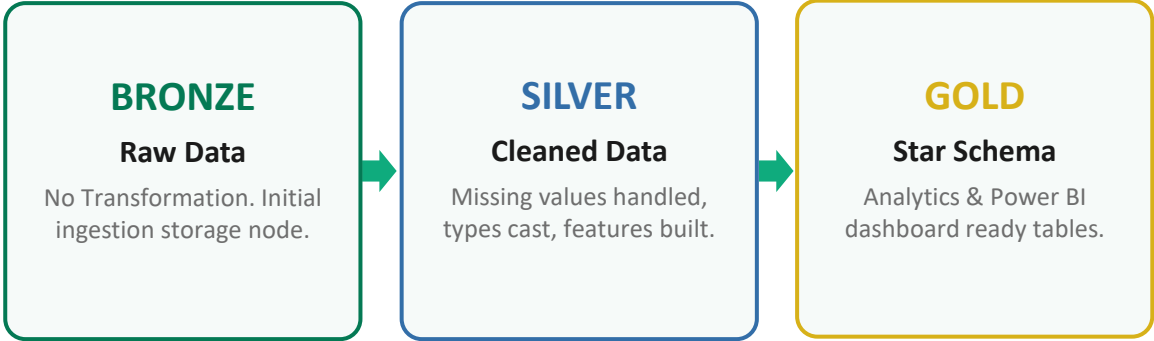
Responsive Release: Trigger mechanisms.

Data Structure

Dataset Overview

Property	Value
Source	Zenodo – Drug Delivery Dataset
Data Type	CSV (Comma Separated Values)
Number of Rows	2,374
Number of Columns	25

Data Flow (Medallion Architecture)



Programming Languages + Frameworks

Languages

Python: Main logic processing runtime.

ML & Frameworks

Pandas: Memory tabular modeling.

NumPy: Fast array processing.

Matplotlib/Seaborn: Clinical plots.

Scikit-Learn: GroupKFold, Nested Cross-Validation & RandomizedSearchCV.

XGBoost + MultiOutputClassifier: Hyperparameter fine-tuned multi-target classification.

Platforms

Excel/Figma: Initial scoping & prototyping.

Google Colab: Development scripting notebook.

Power BI: Interactive presentation.

Storage Format

Excel (.xlsx): Standard data export format.

CSV Files: Raw registry files.

Live Application + Test

✓ STATUS: PRODUCTION READY

Power BI Dashboard — 10 Pages: The interactive Power BI dashboard was built using the Gold Layer data model and provides comprehensive insights into nanoparticle biodistribution. It includes KPI cards, interactive charts, slicers, treemaps, scatter plots, and drill-down analysis.

Page	Description
P1 P2	Home Dashboard General KPI Overview
P3 P4	Nanoparticle Design Targeting Strategy
P5 P6	Tumor Biology Surface Coating (PEG)
P7 P8	Time & Dosage Analysis Responsive Release
P9 P10	Organic vs Inorganic Machine Learning Prediction

Testing & Validation

Data Validation

Verified data quality after cleaning. Missing values handled and data consistency confirmed.

Pipeline Validation

Bronze → Silver → Gold pipeline executed successfully. Star Schema relationships validated for Power BI.

Machine Learning Validation

Hyperparameter Fine-Tuned Multi-Output XGBoost model validated using Nested GroupKFold Cross-Validation. Performance evaluated using Accuracy, Precision, Recall, and F1-Score.

Project Deliverables

Data Pipeline

Bronze Layer: Raw Data Ingestion (CSV).

Silver Layer: Data Cleaning & transformed data (CSV).

Gold Layer: Star Schema (1 Fact + 7 Dimensions).

Analytics & Visualization

EDA (Python): Exploratory data analysis

Figma Design: UI/UX dashboard design

Power BI Dashboard: Interactive biodistribution insights

Machine Learning

Multi-Output XGBoost Model: Hyperparameter Fine-Tuned Multi-Output XGBoost Model: AI classification model

Class Prediction: Tumor_%ID & Selectivity_Index Prediction

Feature Importance Analysis: Most influential nanoparticle features

Streamlit Web Application: Real-time nanoparticle performance prediction

Documentation

Ch 1: Project Overview

Ch 2: Technical Architecture & Data Engineering

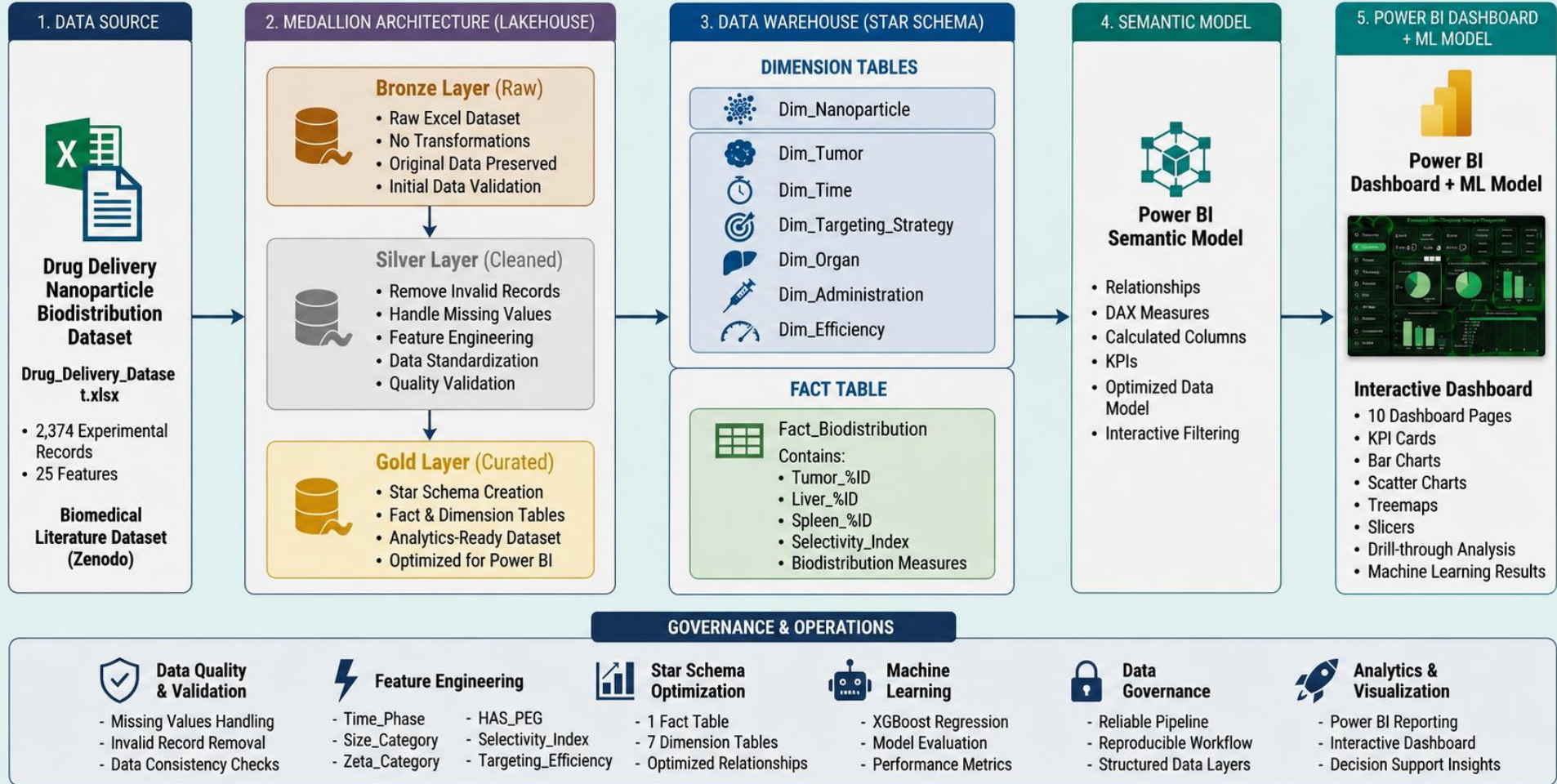
Ch 3: Exploratory Data Analysis (EDA)

Ch 4: Machine Learning

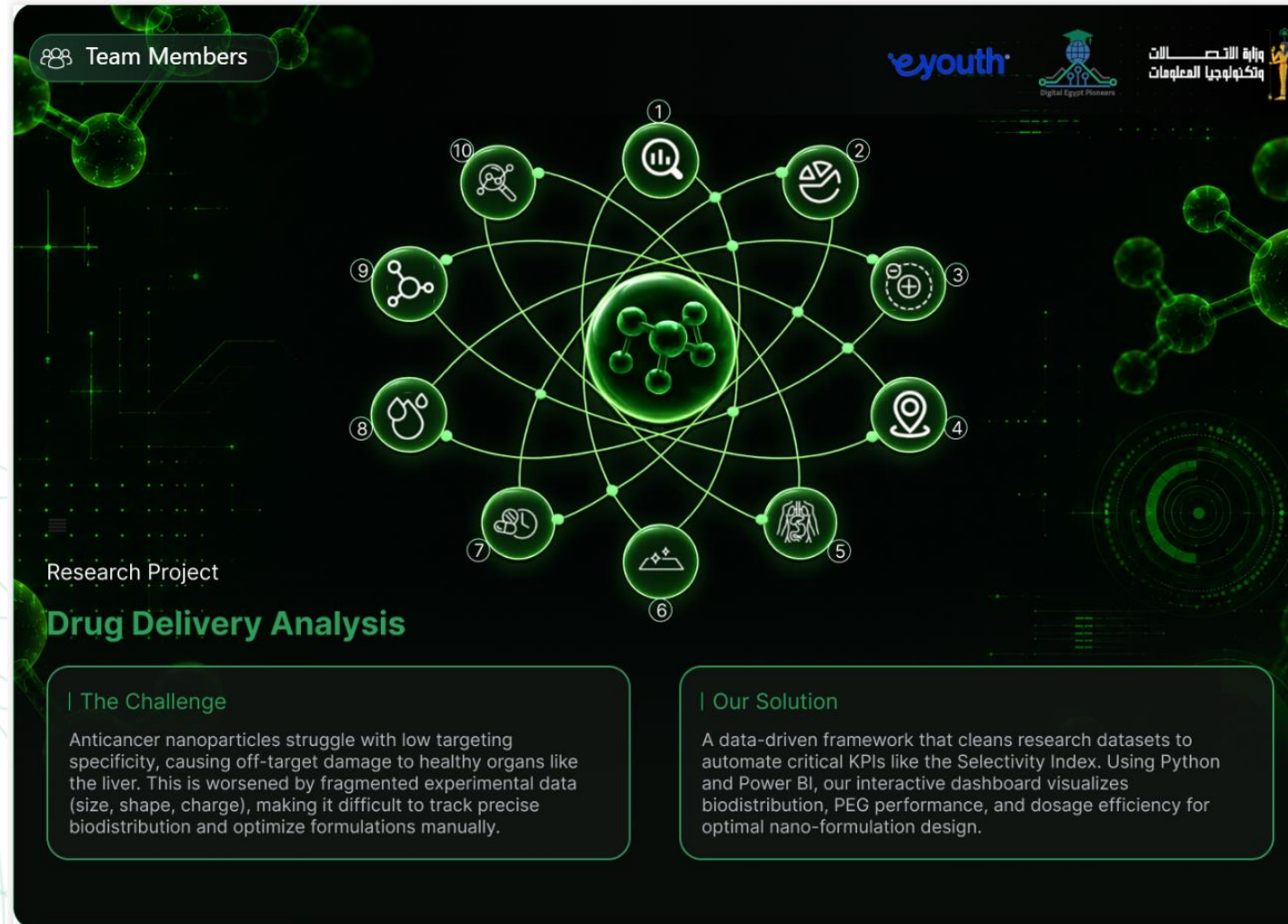
Ch 5: Dashboard & Figma design

System Architecture

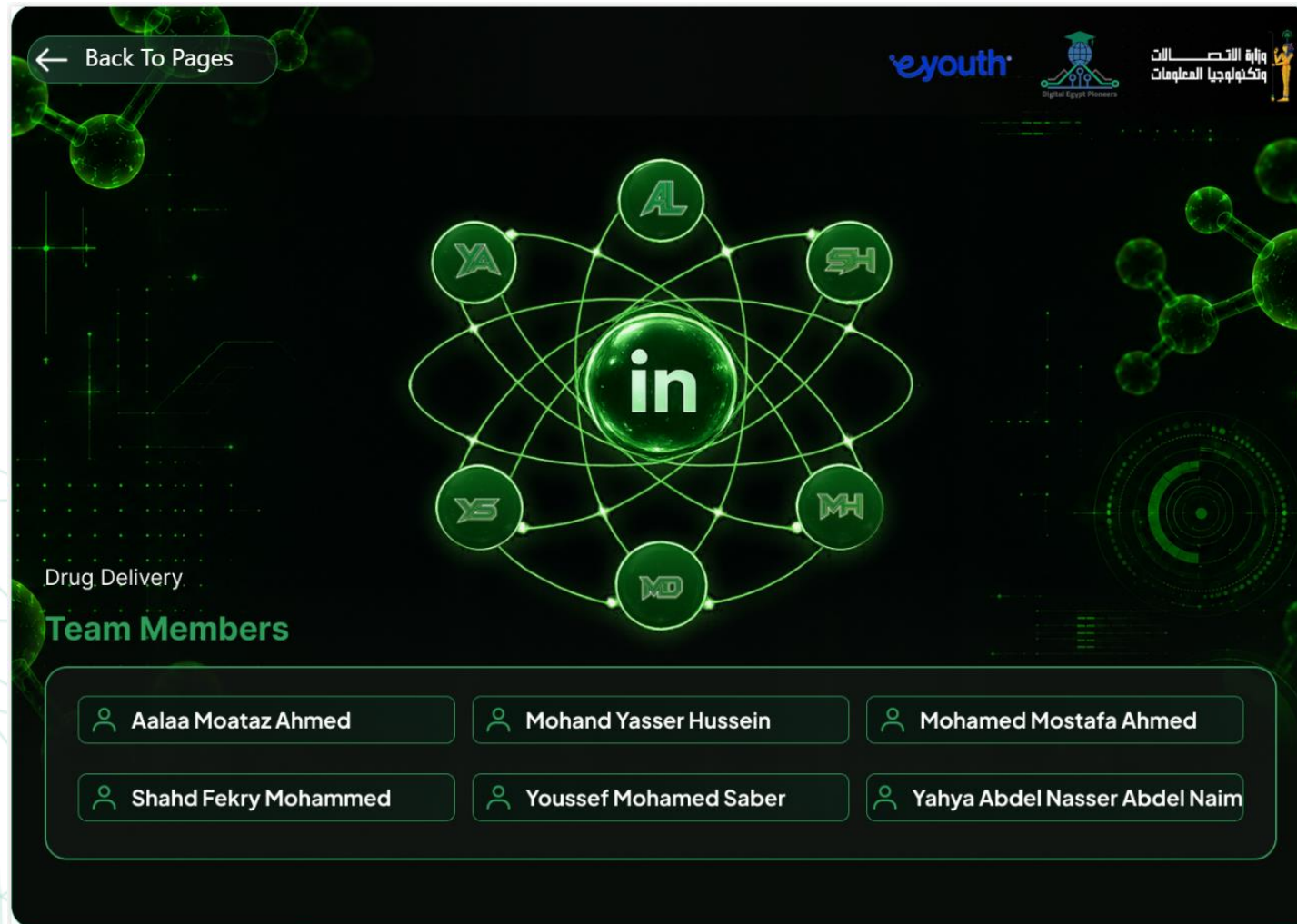
Drug Delivery Data Pipeline – Microsoft Fabric End-to-End Architecture



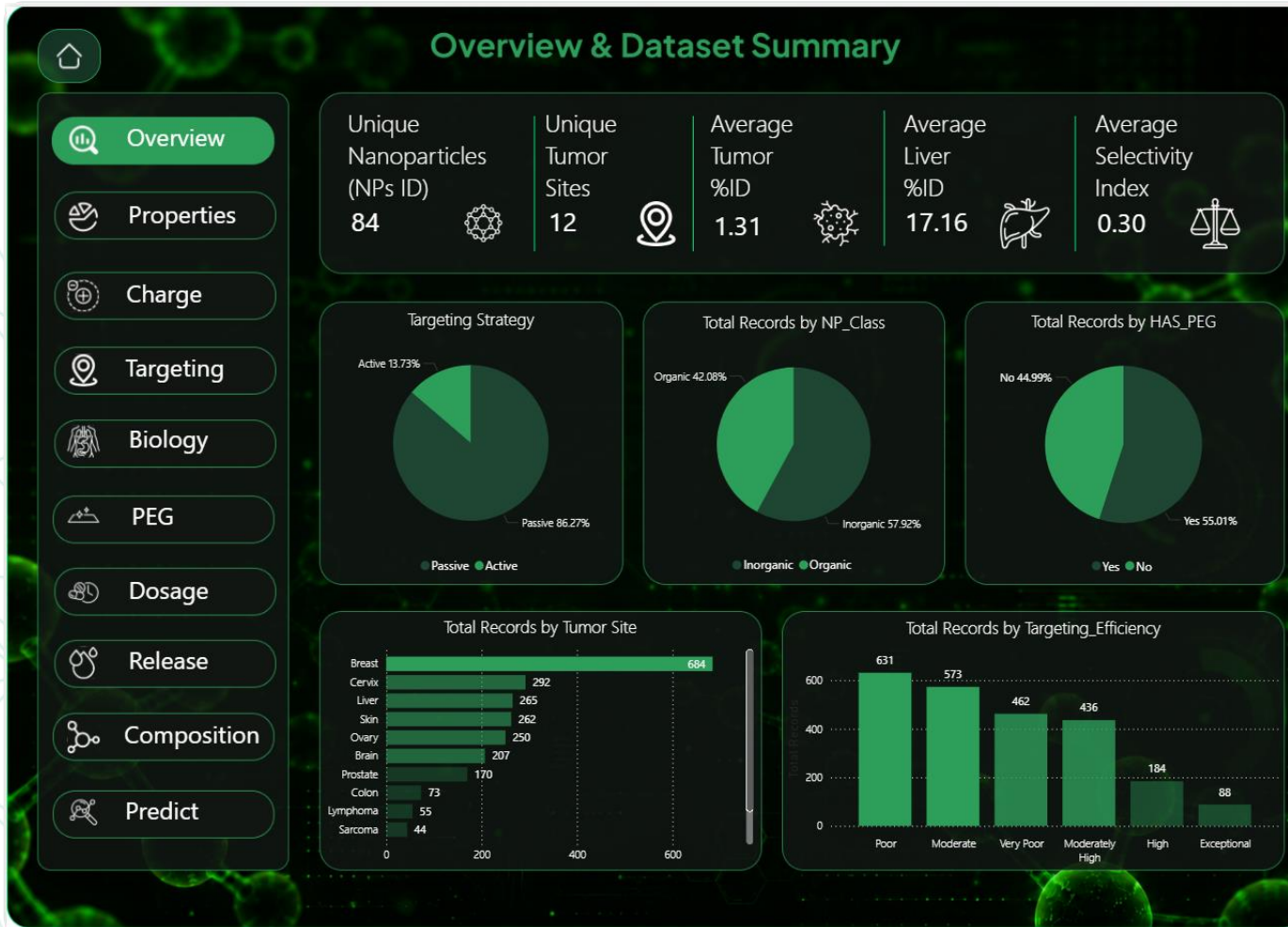
Power BI Dashboard: Landing Page



Power BI Dashboard: Team Members Page



Overview



P1 — Overview

Presents a general summary of the dataset, including experiment distribution, nanoparticle classes, tumor sites, organs, and overall biodistribution statistics.

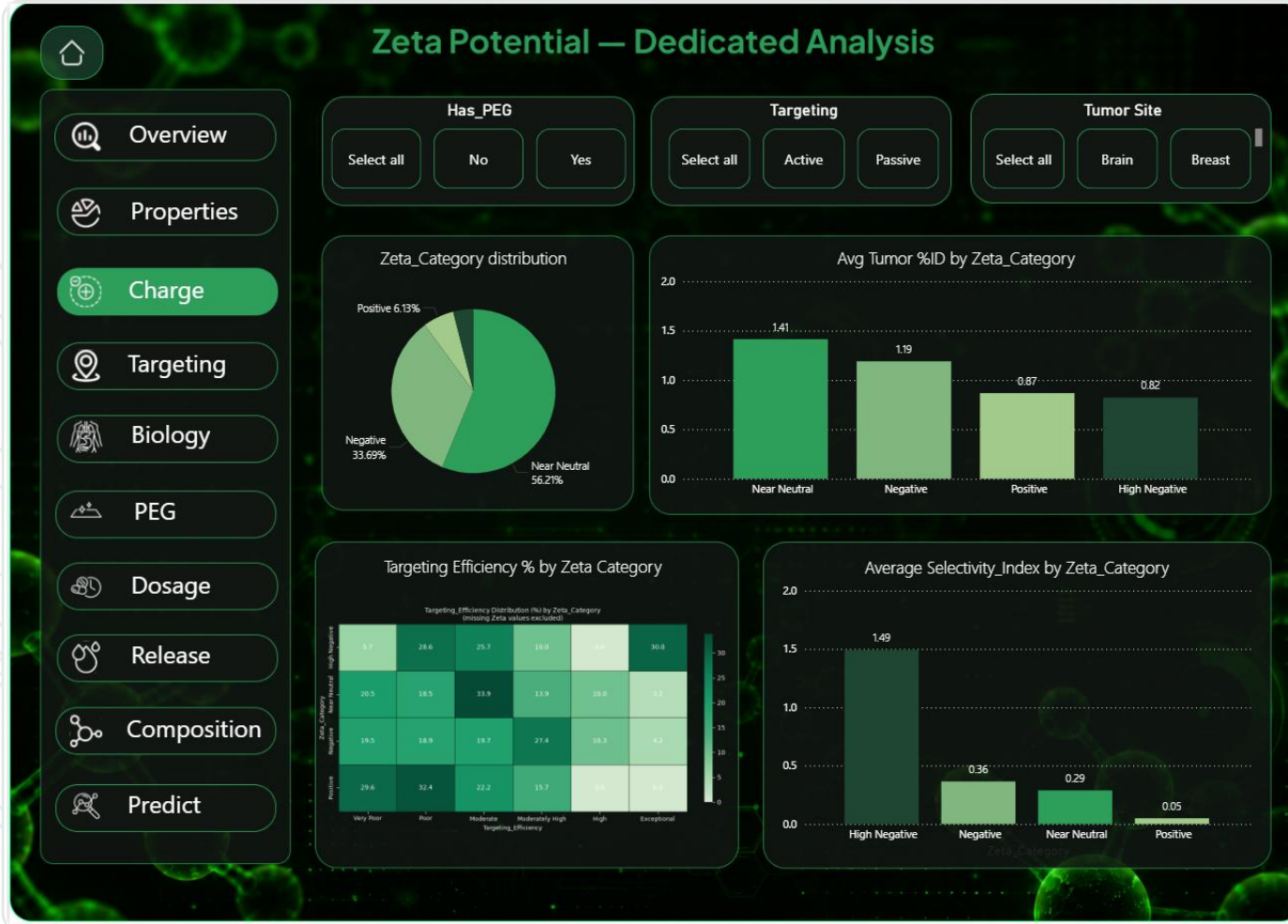
Nanoparticle Design



P2 — Nanoparticle Design

Analyzes how nanoparticle properties such as size, shape, core material, and surface charge influence Tumor_%ID and Targeting Efficiency.

Zeta Potential (Charge)



P3 — Zeta Potential – (Charge)

A dedicated view of nanoparticle surface charge (zeta potential), showing how charge category relates to Tumor_%ID, Targeting Efficiency, and Selectivity_Index.

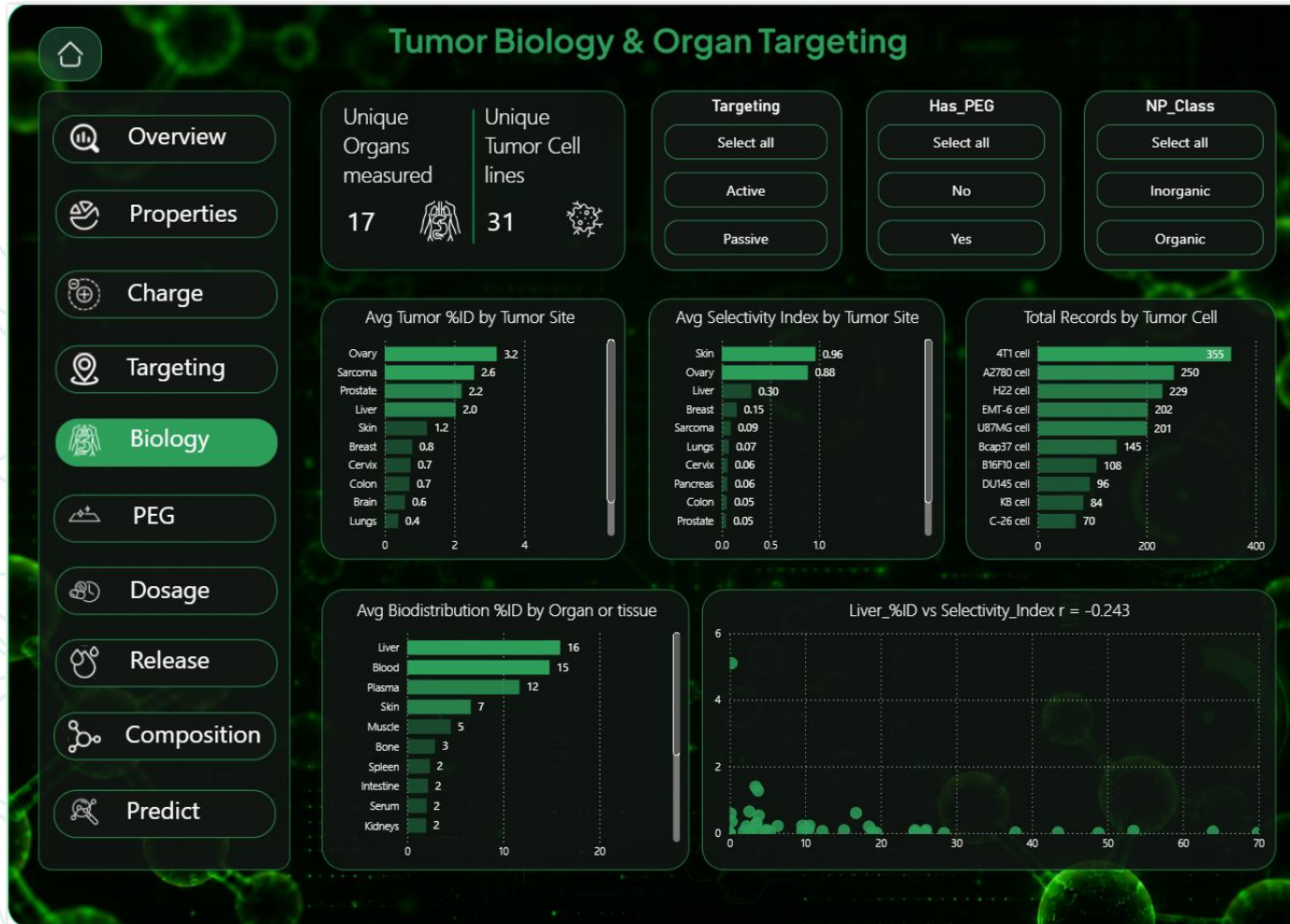
Targeting Strategy



P4 — Targeting Strategy

Compares active and passive targeting approaches to evaluate their impact on Tumor_%ID, Selectivity_Index, and overall delivery performance.

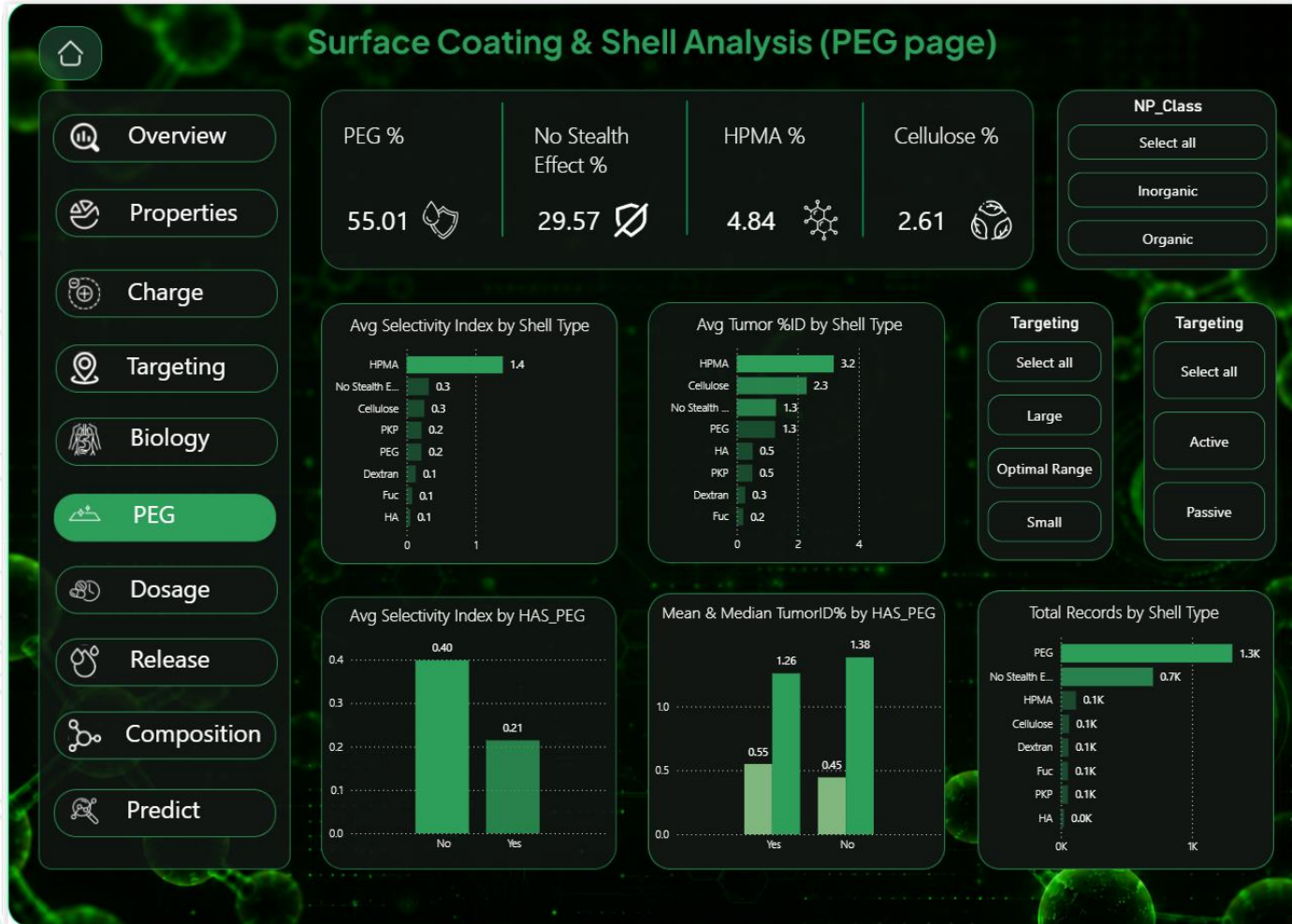
Tumor Biology



P5 — Tumor Biology

Explores biodistribution across different tumor sites, tumor cell lines, and organs to identify patterns in drug accumulation.

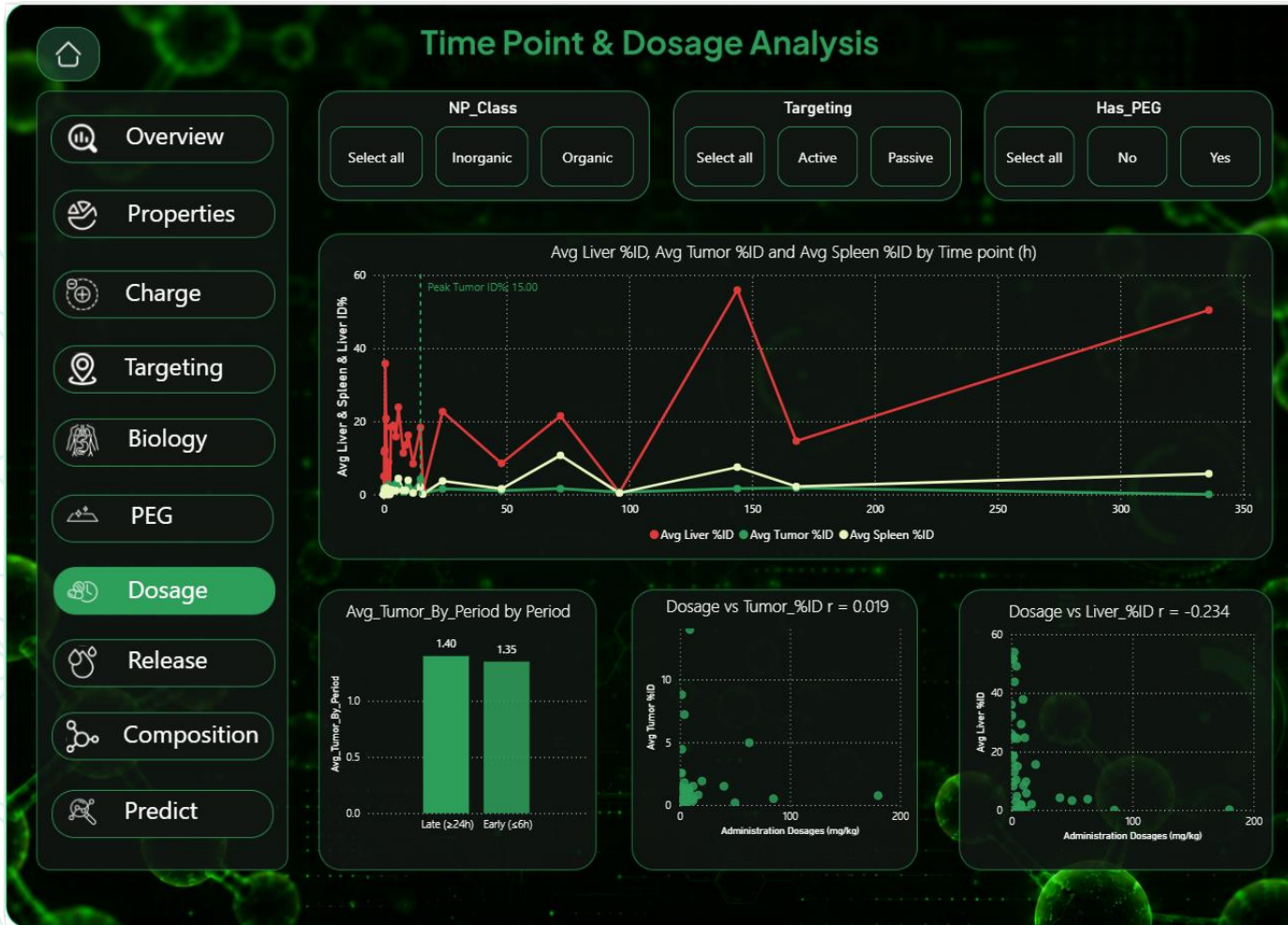
Surface Coating (PEG)



P6 — Surface Coating (PEG)

Evaluates the effect of PEGylation and shell materials on nanoparticle biodistribution, circulation time, and targeting selectivity.

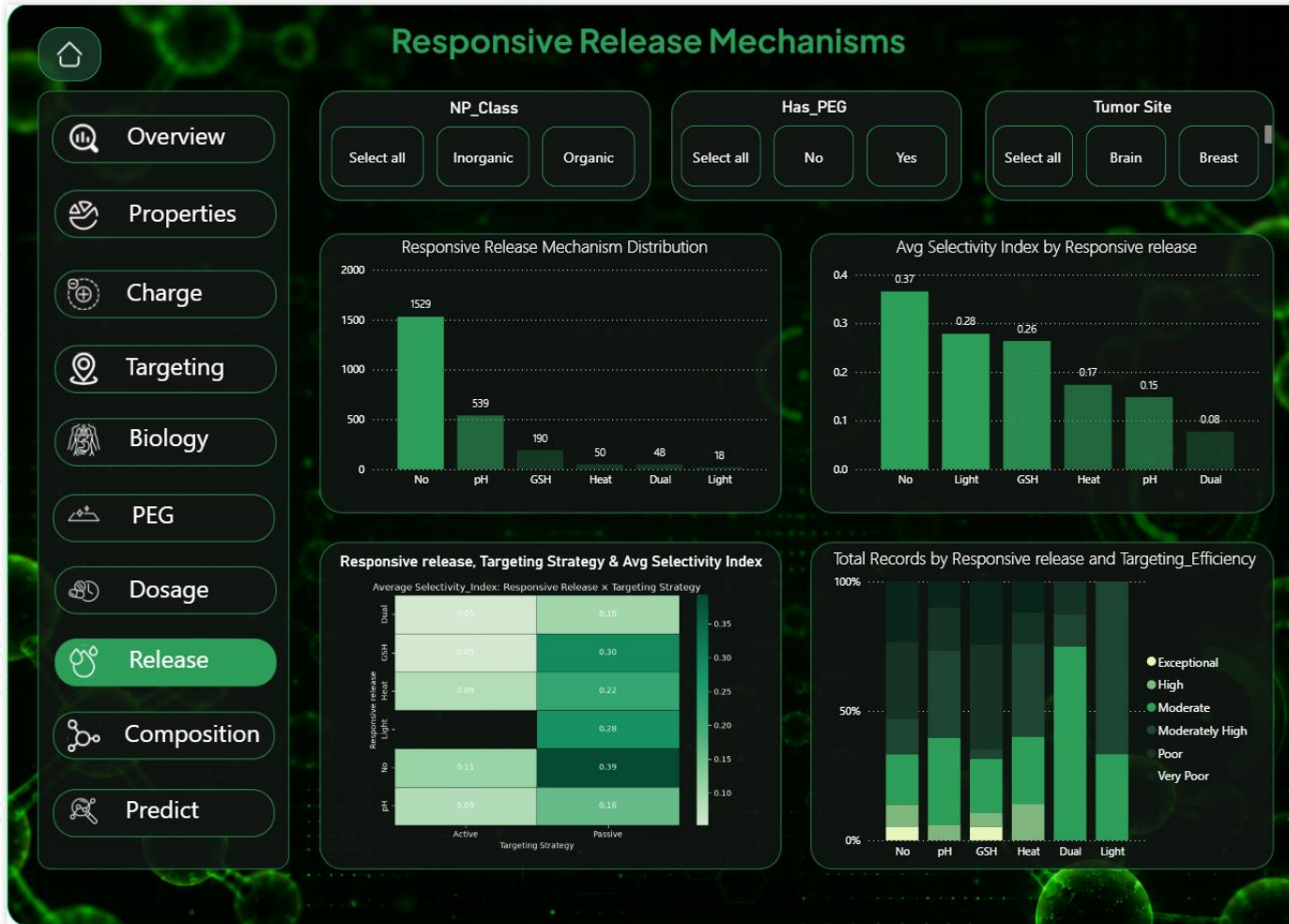
Time & Dosage Analysis



P7 — Time & Dosage Analysis

Examines how administration time points and dosage levels affect drug distribution, Tumor_%ID, and treatment efficiency.

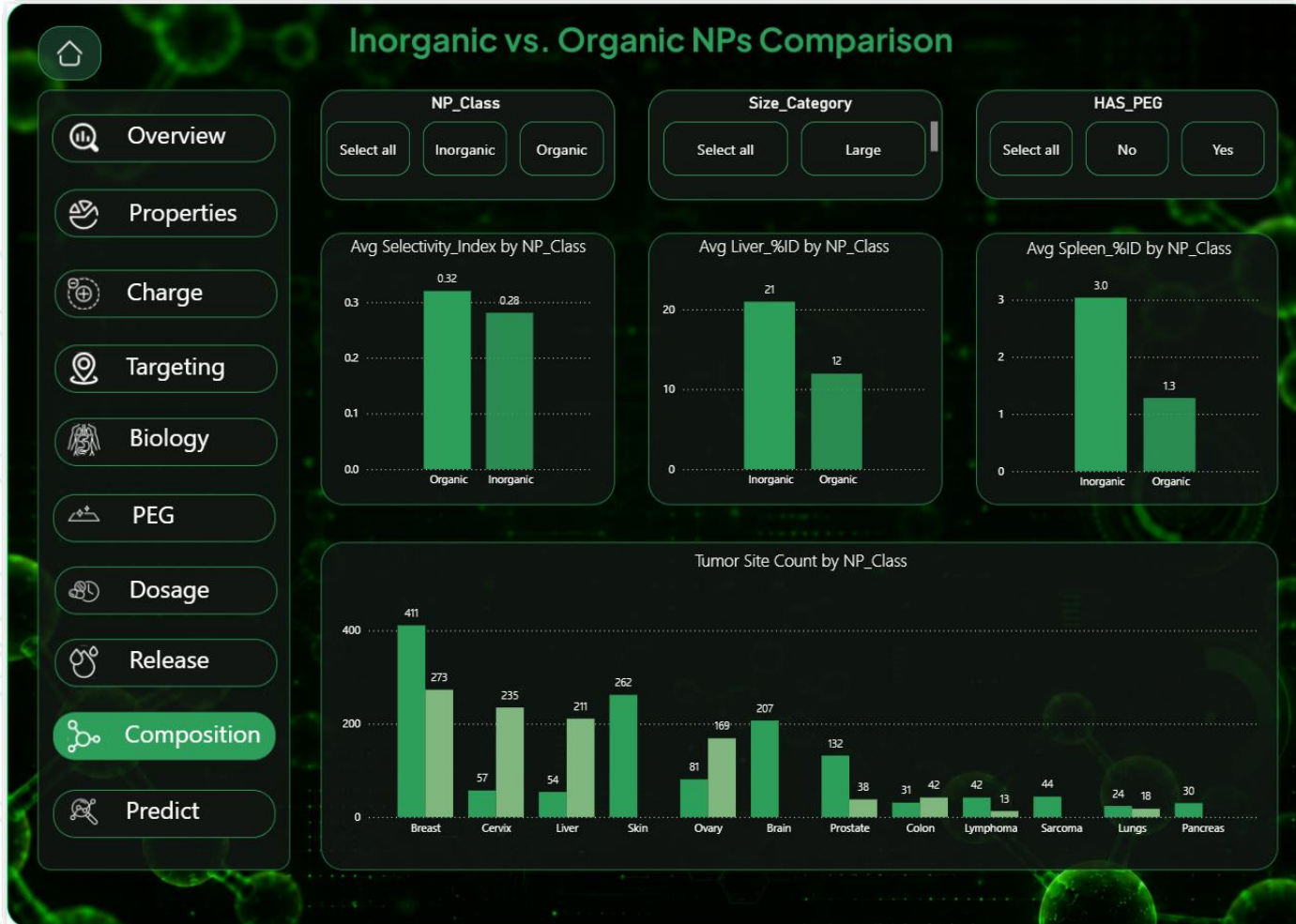
Responsive Release



P8 — Responsive Release

Compares different responsive release mechanisms (such as pH, GSH, Heat, and Light) to determine their influence on Targeting Efficiency and Selectivity_Index.

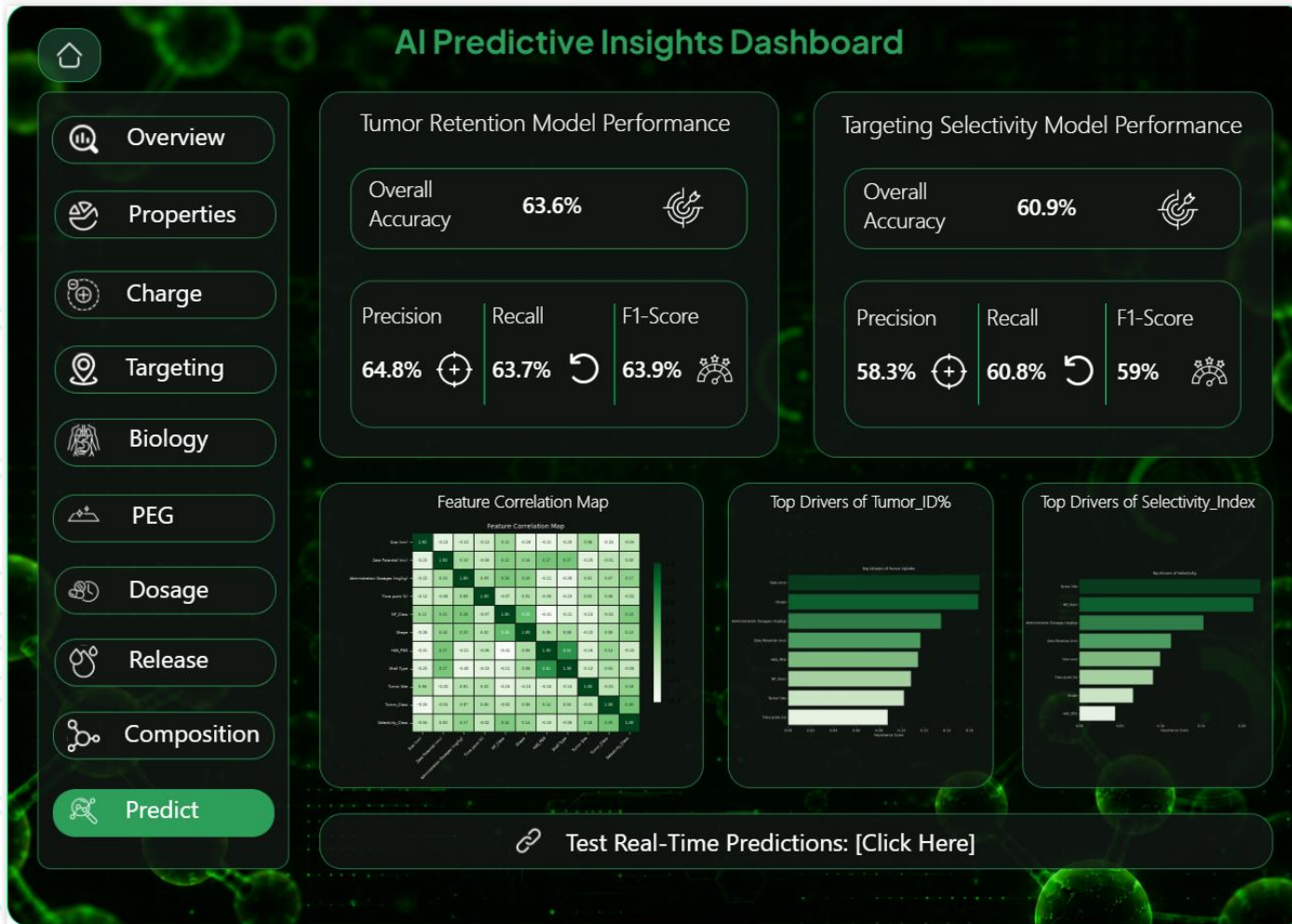
Organic vs Inorganic Nanoparticles



P9 — Organic vs Inorganic

Compares organic and inorganic nanoparticles in terms of biodistribution, Tumor_%ID, Liver_%ID, and Targeting Efficiency.

AI Predictive Insights



P10 — AI Predictive Insights Dashboard

Evaluates the Hyperparameter Fine-Tuned Multi-Output XGBoost model, visualizes classification performance and feature importance, and enables real-time prediction of Tumor_%ID and Selectivity_Index for new nanoparticle formulations.

Project Team + Roles

Team Leader: **Shahd Fekry**

Shahd Fekry

Excel
EDA
Power BI
Documentation
Project Demo

Yahya Abdel Nasser

Excel
Medallion Architecture
EDA
Figma Design
Power BI
Documentation

Aalaa Moataz

Excel
Medallion Architecture
EDA
Figma Design
Machine Learning
Documentation

Mohand Yasser

Excel
Medallion Architecture
EDA
Presentation
Project Demo

Mohamed Mostafa

Excel
Presentation

Youssef Mohamed

Excel
Power BI

Thank You

Contact Information

✉ Shahd Fekry

✉ Yahya Abdel Nasser

✉ Aalaa Moataz

✉ Mohand Yasser

✉ Mohamed Mostafa

✉ Youssef Mohamed

Invitation for questions or feedback